

MAHAK BANSAL

Full Stack Engineer

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SUMMARY

Full Stack Engineer with 3+ years of experience building scalable, production-grade web applications. Specializes in React.js, Next.js, Node.js, TypeScript, and AWS Serverless architecture. Proven track record engineering high-performance CRM platforms, real-time streaming systems, automated lead-ingestion pipelines, and multi-gateway payment infrastructure. Strong expertise in MongoDB, Redis, REST API design, and API integration, with hands-on experience integrating third-party APIs including WhatsApp Business, Meta Ads, Google Ads, Razorpay, and Stripe. Experienced with CI/CD pipelines, Agile/Scrum workflows, version control with Git, and collaborating across cross-functional teams to ship features on tight deadlines. Delivers measurable business impact by reducing manual workflows, improving conversion rates, and optimizing system performance at scale.

TECHNICAL SKILLS

Frontend	React.js, Next.js, TypeScript, JavaScript, HTML5, CSS3, Redux Toolkit, RTK Query, TanStack Query, Flutter, Dart
Backend	Node.js, Express.js, REST API Design, API Integration, JWT Authentication, Serverless Architecture, Microservices
Databases	MongoDB, MySQL, SQL, Firebase, Firestore
Cloud & DevOps	AWS Lambda, Amazon API Gateway, CloudFront, GCP, Git, GitHub, CI/CD Pipelines, Agile, Scrum, Version Control
Real-Time	Redis, WebSockets, BullMQ, Worker Threads, Agora SDK
Integrations	WhatsApp Business API, Meta Ads API, Google Ads API, Razorpay, Cashfree, Stripe, RevenueCat
AI & Tools	LLM Agents, GitHub Copilot, Prompt Engineering, Automated Scaffolding

WORK EXPERIENCE

Full Stack Engineer | CodnestX Tech Dynamics Pvt. Ltd.
India

May 2024 - Apr 2026 · Ahmedabad,

- Engineered a production **React.js CRM admin dashboard** from scratch with role-based access control (**RBAC**) for 4 plus user types, dynamic lead pipeline views, real-time status updates, and bulk-action workflows - reducing agent task-switching by **40 percent**.
- Built a **reusable React component library** including forms, modals, data tables, filters, and notification panels - cutting UI development time by **35 percent** and enforcing a consistent design system across the platform.
- Optimized React rendering performance using **memoization, lazy loading, and code splitting** - achieving **sub-2 second initial load** on the CRM dashboard under real-world network conditions.
- Implemented state management with **Redux Toolkit** RTK Query and **TanStack Query** for **server state**, caching, background refetching, and loading and error handling across the entire CRM frontend.
- Engineered a **real-time lead-ingestion pipeline** automating **100 percent** of ad-generated lead capture from Facebook and Google Ads - reducing data lag to **under 5 seconds** with zero manual CRM entry.
- Integrated **WhatsApp Business API** with automated follow-up drip sequences, improving sales response time by **70 percent** and increasing customer conversion touchpoints.
- Architected a unified payment gateway layer covering **Razorpay, Cashfree, and Stripe** with **webhook-driven reconciliation** - achieved **zero downtime** post-launch across all payment flows.
- Built production live-streaming infrastructure using **Agora SDK** and Node.js with **token-based session management**, delivering **sub-second latency** for concurrent multi-user video sessions.
- Designed **Redis Pub/Sub** event architecture decoupling real-time notification delivery from core API paths, enabling **horizontal scaling** without service-level code changes.
- Implemented **BullMQ** and **Worker Threads** for bulk notification dispatch with retry logic and **zero message loss** under peak load.
- Architected and managed multiple **AWS Lambda** functions following a **microservices** pattern - each function scoped to a **single responsibility** including webhook processing and background removal via third-party API - all interconnected through **Amazon API Gateway** and **CloudFront** for unified routing, low-latency delivery, and seamless scalability.

- Accelerated delivery timelines by **25 percent** by integrating **AI coding assistants** and agent-based automation, reducing boilerplate generation and review cycles.

MERN Stack Developer | CleverPush

May 2023 - Jan 2024 · Ahmedabad, India

- Built and maintained **React.js frontend modules** with **reusable**, composable components and clean separation of UI concerns across the application layer.
- Improved UI responsiveness by auditing and refactoring React component trees, **eliminating unnecessary re-renders** and reducing interaction latency in production.
- Engineered secure RESTful APIs with **JWT authentication**, **input validation**, **rate limiting**, and secure header enforcement - following industry-standard API security practices across all endpoints.
- Optimized offline data reliability using advanced **IndexedDB** storage strategies, achieving **zero data loss** during network interruptions across all supported browsers.

Flutter and MERN Stack Developer | AppsDeployer Tech Services and Solutions May 2022 - Apr 2023 · Indore, India

- Re-engineered the **React.js frontend** of a stalled **MERN stack** product - rebuilt component architecture, fixed broken **API integrations**, and stabilized state management to deliver a production-ready UI within a tight deadline.
- Reduced user-facing errors by **40 percent** by diagnosing React state mismatches and Node.js API defects, dramatically improving platform stability and earning full client sign-off.
- Delivered **50 percent efficiency improvement** by revamping a legacy **Flutter** Android app with new UI features and performance-optimized rendering, confirmed by post-launch client metrics.

PROJECTS

Real Estate CRM - Lead-to-Conversion Platform — React.js, Node.js, MongoDB, Redis, AWS Lambda, WhatsApp API, Razorpay

- Built end-to-end from scratch: **automated ad-lead ingestion** from Meta and Google Ads, **RBAC** for 4 plus user roles, WhatsApp drip sequences, and **multi-gateway payment infrastructure**.
- Cut manual data entry by **100 percent**, reduced follow-up time by **70 percent**; Redis and AWS Lambda maintain API response times **under 200ms** at peak load.

Live Streaming SaaS Module — Node.js, Agora SDK, MongoDB, JWT

- Engineered a production live-streaming feature with **token-based session management**, multi-user concurrency, and a Node.js orchestration layer delivering **sub-second video latency** under real-world load.

Tech Solutions B2C Marketplace — React.js, Node.js, MongoDB, Firebase

- Independently architected a **customer-to-vendor matching** platform with smart filtering, **Firebase authentication**, and a full admin panel for inventory, user management, and order lifecycle.

Cross-Platform Object Capture App — Flutter, Firebase

- Sole developer of a **cross-platform mobile app** with Google and email OAuth, real-time object scanning, and **offline-tolerant Firebase sync** - from design to production deployment.

Agriculture Export Website — HTML, CSS, Node.js

- **SEO-optimized** production website for an international agricultural exporter supporting buyer acquisition across multiple markets. Live: devagroexim.com

EDUCATION

B.Tech — Computer Science & Engineering · Mandsaur University

2018 – 2022 · Mandsaur, MP

CGPA: 8.5 / 10

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Software Engineering

LANGUAGES & INTERESTS

Languages: Hindi (Native) · English (Full Professional Proficiency)

Interests: AI & Emerging Technologies · System Design · Open Source Contribution · Music · Travelling